

Institute of Actuaries of India

**Subject ST6 – Finance and
Investment B**

October 2015 Examination

INDICATIVE SOLUTION

Solution 1 :

i) A total return swap is a means of transferring the total economic exposure, including both market and credit risk, of the underlying asset. The payer of a total return swap can confidentially remove all the economic exposure of the asset without having to sell it.

bond forward or bond futures contract is an agreement whereby the short position agrees to deliver pre-specified bonds to the long at a set price and within a certain time window.

The forward contract is an agreement between two counterparties to exchange bonds at an agreed price and time in the future. [3]

(ii) Swap spread = Swap rate – Gilt rate

Buy forward to enter in receiver total return swap at $t=1$ and forward to sell a bond of the same duration.

Then there would be following cashflow

$$\begin{aligned} \text{Swap cashflow} &= \sum N * (i_{\text{strike}} - i_{t=1}) \\ \text{Bond cashflow} &= \sum N * (Govn_{t=1} - Govn_{\text{strike}}) \\ i_{\text{strike}} &= Gov_{\text{strike}} + \text{spread}_{\text{strike}} \\ i_{t=1} &= Gov_{t=1} + \text{spread}_{t=1} \end{aligned}$$

Combining above equation we get the total cashflow as

$$\text{Combined Cashflow} = \sum A * (\text{spread}_{\text{strike}} - \text{spread}_{t=1})$$

Thus when the swap spread goes down the company make the profits which compensates for the loss on the portfolio effectively acting as hedge on the portfolio. [8]

iii) Forward contracts are subject to counterparty risk, which is the risk that the party on the other side of the trade defaults on their contractual obligation

Given this is forward contract there is effectively a risk of making unlimited loss if the swap spread rises and if the rise is not compensated by the movement in the liability of the insurance company then it may effectively results in default. [4]

[15 Marks]

Solution 2**(i) Fixed rate exposure**Current value of Bond = $1,00 / (1+r+rp)^5 = 71.3\text{lakh}$ Capital on bond = $5\% * 71.3 = 3.6\text{lakh}$

option

value of option using BS formula

Input Data	
Stock Price now (P)	8000
Exercise Price of Option (EX)	8000
Number of periods to Exercise in years (t)	5
Compounded Risk-Free Interest Rate (rf)	5.00%
Standard Deviation (annualized σ)	25.00%

Output Data	
Present Value of Exercise Price (PV(EX))	6230.4063
$\sigma * t^{.5}$	0.5590
d1	0.7267
d2	0.1677
Delta N(d1) Normal Cumulative Density Function	0.7663
Bank Loan N(d2)*PV(EX)	3530.1006

Value of Call	2600.3146
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Number of option	1250
Value of option	3,250,393.20

The option value would be lease in scenario would low volatility and low equity level hence

Input Data	
Stock Price now (P)	6000
Exercise Price of Option (EX)	8000
Number of periods to Exercise in years (t)	5
Compounded Risk-Free Interest Rate (rf)	5.00%
Standard Deviation (annualized σ)	10.00%

Output Data	
Present Value of Exercise Price (PV(EX))	6230.4063
$\sigma * t^{.5}$	0.2236
d1	-0.0567
d2	-0.2803
Delta N(d1) Normal Cumulative Density Function	0.4774
Bank Loan N(d2)*PV(EX)	2427.4594

Value of Call	436.8550
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Number of option	1250
Value of option	546,068.72

Loss in the derivative value: 27.04lk

Total capital requirement

total capital requirement bond	356,493.09
Option	2,704,324.48
Sum	3,060,817.56

Thus total capital requirement for the insurance would be 30lk. [8]

ii) The cost of holding 30.6lk would be $8.5 \times 5 \times 30.6$ that is 4.6lk. Allowing for the cost of capital the asset is not very attractive as option to earn excess return or gain equity exposure.

[3]

[11 Marks]

Solution 3:

(i) The Merton model takes an overly simple debt structure, and assumes that the total value A_t of a firm's assets follows a geometric Brownian motion under the physical measure $dA_t = \mu A_t dt + \sigma A_t dW_t$, $A_0 > 0$,

where μ is the mean rate of return on the assets and σ is the asset volatility.

We also need further assumptions: there are no bankruptcy charges, meaning the liquidation value equals the firm value; the debt and equity are frictionless tradeable assets.

Large and medium cap firms are funded by shares ("equity") and bonds ("debt"). The Merton model assumes that debt consists of a single outstanding bond with face value K and maturity T . At maturity, if the total value of the assets is greater than the debt, the latter is paid in full and the remainder is distributed among shareholders. However, if $A_T < K$ then default is deemed to occur: the bondholders exercise a debt covenant giving them the right to liquidate the firm and receive the liquidation value (equal to the total firm value since there are no bankruptcy costs) in lieu of the debt. Shareholders receive nothing in this case, but by the principle of limited liability are not required to inject any additional funds to pay for the debt.

[5]

ii)

Using Merton Model**Data**

Value of firm	100
Face value of debt	60
Maturity (year)	10
Interest rate	5.00%
Volatility	30.00%
Mkt risk premium	6.00%
Beta asset	1

Solution:-

Using Black Scholes:

Step 1: Calculate d1 and d2

$$\begin{aligned} d1 &= 1.538 \\ d2 &= 0.589 \end{aligned}$$

Step 2: Find N(d1) and N(d2)

$$\begin{aligned} N(d1) &= \text{Delta} = 0.9380 \\ N(d2) &= \text{Proba NoDefault} = 0.7221 \\ &\text{(in a risk-neutral world)} \end{aligned}$$

Step 3: Find E using Black-Scholes

$$\text{Market value of equity} = 67.52$$

Characteristic of risky debt

Market value of debt	32.48
Yield to maturity	6.14%
Spread (bp)	<u>114</u>

Decomposition of risky debt

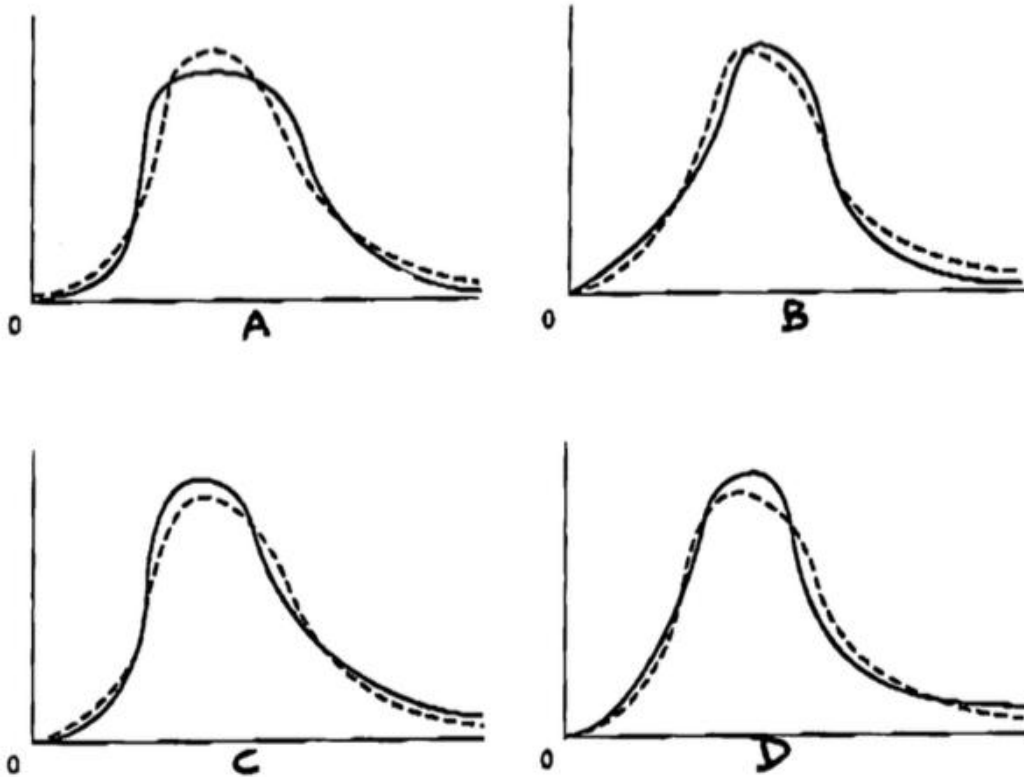
Riskless debt	36.39
Put	3.91

[5]

[10 Marks]

Solution 4:

The probability distribution could have 4 tail behaviors



dotted is lognormal

solid line is actual

A - both tails are lower than lognormal

D - both tails are higher than lognormal

C - Left tail is lower than lognormal and right tail is higher

B - Right tail is higher than lognormal and left tail is lower

The pricing would vary significantly for the out of money option. Fatter is the tail higher is the value of option. Thus BS formula would underprice

A – overprice both out of money call and put option

D- underprice both out of money call and put options.

B – underprice out of money call and overprice out of money put

C- under price out of money call and overprice out of money call

[6 Marks]

Solution 5:

Using the above equation we can obtain the following binomial tree

Interest rate using the above model interest rate would be step change by $\text{Sqrt}(r(t-1)*3)/10$

			3.94
		3.61	
	3.30		3.29
3.00		2.99	
	2.70		2.69
		2.42	
			2.15

The cap consist of 3 options at one at end of each year.

Option 1

	0.3000
0.14556683	
	0.0000

Option 2

		0.614642654
	0.297345234	
0.144278677		0
	0	
		0

Option 3 :

			0.943943853
		0.592822259	
	0.35377888		0.285341455
0.204362102		0.138474445	
	0.067392829		0
		0	
			0

Caplet price is

$0.49\% * 1\text{cr}$

49K

[8 Marks]

Solution6:

Using the expected return and volatility for the first security, the market price of risk for Nifty is:

$$\frac{\mu_1 - r}{\sigma_1} = \frac{0.20 - 0.06}{0.25} = 0.56$$

When there is just one source of randomness, derivatives on the same underlying asset would have the same market price of risk; i.e.

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2} = 0.56$$

$$\Rightarrow \mu_2 = (0.56 * 0.35) + 0.06 = 0.256$$

Thus, the expected return from the second security = 25.6%

[6 Marks]

Solution 7:

(i) Let's evaluate both the transactions.

Selling the stock

- Receives Rs. $(650 * (1-1\%)) = \text{Rs. } 643.50$
- Loss of dividend of Rs. 45 as on 15th October 2015 whose value as on 1st October 2015 is
 - Rs. 44.90 (if computed @ 5% for 15 days)
 - Rs. 44.89 (if computed @ 6% for 15 days)

Since the client will be net lender rather than borrower the applicable rate of interest is 5%. And hence the loss of dividend in present value terms should be Rs. 44.90

Buying the futures

- Pay the brokerage: Rs. $(594 * 0.005) = \text{Rs. } 2.97$
- Pay the margin: Rs. $(651 * 0.05) = \text{Rs. } 32.55$
- Pay Rs. $(594 - 32.55) = \text{Rs. } 561.45$ on maturity (31st October 2015) and get the shares.
- The value of this amount as on 1st October 2015 is the present value of Rs. 561.45 @ 5% for 31days = Rs. 559.08

In the nutshell, the expected gain from the strategy today is:

$$\text{Rs. } (643.50 - \text{Rs. } 44.90 - \text{Rs. } 2.97 - \text{Rs. } 32.55 - \text{Rs. } 559.08) = \text{Rs. } 4.00$$

Hence, the client can be advised to go ahead with the strategy

[9]

ii) The risks in implementing this strategy are:

- The discount is not as substantial as you may infer by looking at the prices.
- Even if the prices are known for calculating the gain/loss before putting the order on the exchange, the actual transacted price may be different compared to the price used in calculating the gains.

- The company may either reduce or increase the amount of dividend after the transactions. The decrease is beneficial under the strategy but not the increase.
- Taxes have been ignored in the above calculations and they may decrease/increase the benefits.
- The daily settlements have been ignored in the above calculations. This may give rise to further cash-flow management problems and interest gain/losses on the same.

[5]

[14 Marks]**Solution 8:****(i):** The process followed by B, the bond price, is from Ito's lemma:

$$dB = \left[\frac{\partial B}{\partial y} \mu(\alpha - y) + \frac{\partial B}{\partial t} + \frac{1}{2} \frac{\partial^2 B}{\partial y^2} \sigma^2 y^2 \right] dt + \frac{\partial B}{\partial y} \sigma y dZ$$

Also,

$$B = \int_0^\infty 2e^{-yt} dt = \frac{2}{y}$$

Thus,

$$\frac{\partial B}{\partial t} = 0; \frac{\partial B}{\partial y} = -\frac{2}{y^2}; \frac{\partial^2 B}{\partial y^2} = \frac{4}{y^3}$$

This implies

$$dB = \left[-\mu(\alpha - y) \frac{2}{y^2} + \frac{2\sigma^2}{y} \right] dt - \frac{2\sigma}{y} dZ$$

[6]

(ii) The expected instantaneous rate at which capital gains are earned from the bond is:

$$\left[-\mu(\alpha - y) \frac{2}{y^2} + \frac{2\sigma^2}{y} \right]$$

The expected interest per unit time is Rs. 2. The total expected instantaneous return is:

$$\left[2 - \mu(\alpha - y) \frac{2}{y^2} + \frac{2\sigma^2}{y} \right]$$

When expressed as a percentage of the bond price this is:

$$\frac{\left[2 - \mu(\alpha - y) \frac{2}{y^2} + \frac{2\sigma^2}{y} \right]}{\frac{2}{y}} = y - \frac{\mu}{y}(\alpha - y) + \sigma^2$$

[4]

[10 Marks]

Solution 9:

Payment made by BNP Paribaas on buying the 3 year CDS on \$200 million from DB: $\$(200 * 0.0113)$ million = \$2.26 million

Payment received by BNP Paribaas on selling the 5 year CDS on \$300 million to DB: $\$(300 * 0.0027)$ million = \$0.81 million

Payment received by BNP Paribaas on selling the 1 year CDS on \$100 million to DB: $\$(100 * 0.0043)$ million = \$0.43 million

Net Annual payment made by BNP Paribaas to DB in the first year is $\$(2.26 - 0.81 - 0.43)$ million = \$1.02 million

[5 Marks]

Solution 10:

Note that this is a semiannual payment – hence all annual coupon rates need to be divided by 2.

Payment by RBS is @ $((6.5\%/2) + 2\%) = 5.25\%$

Actual amount thus paid by RBS: \$400 million @ 5.25 % for 1 year = \$21 million

Receipt by RBS is @ $4.5\% / 2 = 2.25\%$

Actual amount thus received by RBS: \$400 million @ 2.25 % for 1 year = \$9 million

Net payment by RBS: \$21 million - \$9 million = \$12 million

[5 Marks]

Solution 11:

The annual marginal PD is

$$d = \frac{1.5\%}{(1 - 0.50)} = 3.00\%$$

Hence, the cumulative PD for five years is

$$d + s_1d + s_2d + s_3d + s_4d = 3.00\% (1 + 0.970 + 0.941 + 0.913 + 0.885) = 14.1\%$$

Where

$$\begin{aligned} s_1 &= (1 - 3.00\%) = 0.970 \\ s_2 &= s_1(1 - 3.00\%) = 0.941 \\ s_3 &= s_2(1 - 3.00\%) = 0.913 \\ s_4 &= s_3(1 - 3.00\%) = 0.885 \end{aligned}$$

The expected number of defaults is thus $100 * 14.1\%$ or approximately 14.

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With a recovery rate of 50%, the expected loss is therefore 7% of the notional. Thus, all tranches up to the 7% point are wiped out.

[6 Marks]

Solution 12:

The price of the bond at time t is

$$P = e^{-Y(T-t)}$$

Thus,

$$\frac{dP}{dy} = -(T-t)e^{-Y(T-t)}$$

From Ito lemma, the volatility of the bond price is:

$$\sigma \frac{dP}{dy} = -\sigma(T-t)e^{-Y(T-t)}$$

This (the volatility of the bond price) tends to zero as t tends to T ; i.e. the bond gets closer and closer to maturity.

[4 Marks]
